

Foundation of Machine learning: From Intelligence to Algorithms, from Data to Decisions

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Intelligence

- IQ (Intelligence Quotient)

- SQ (Social Quotient)

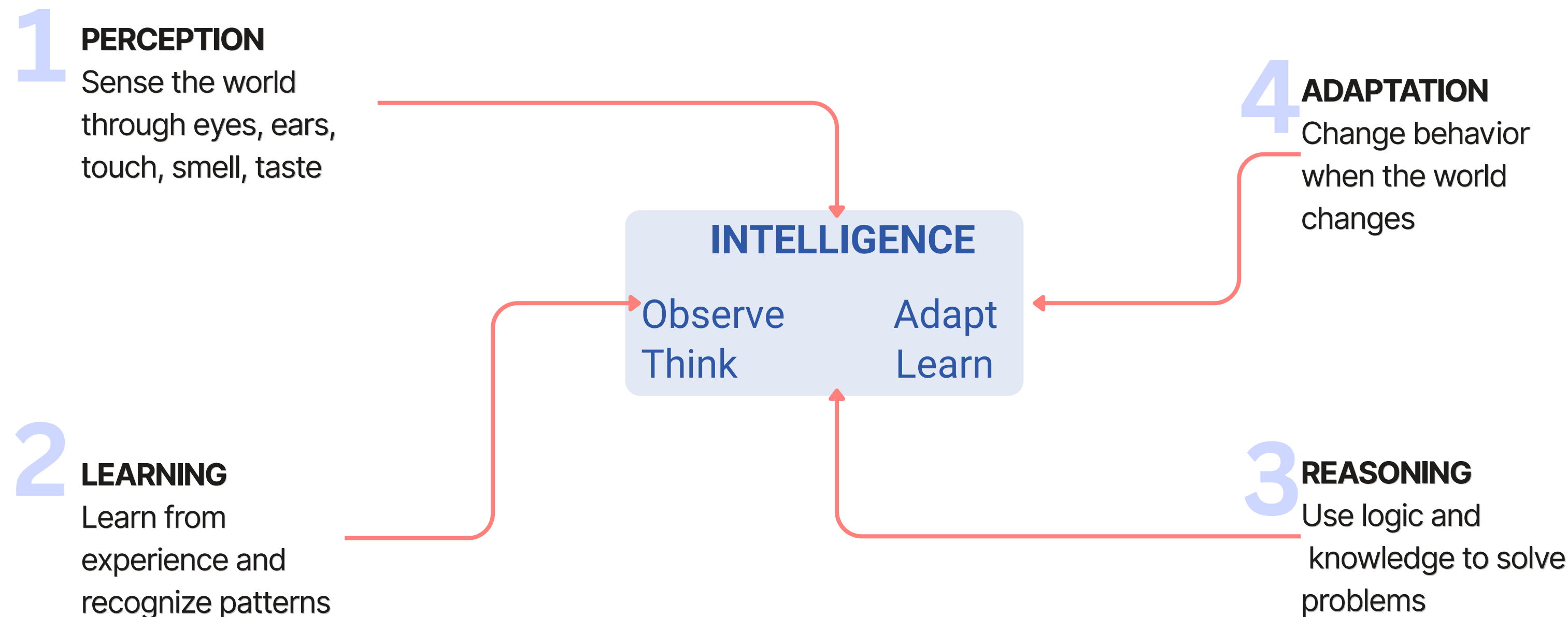
- EQ (Emotional Quotient)

- AQ (Adversity Quotient)



“The measure of intelligence is the ability to change.” – Albert Einstein

Question: What Makes Something Intelligent?



Intelligence is not one ability — it is a system of sensing, learning, thinking, and adapting.

Human Learning vs Machine Learning

HUMAN

- Experience
- Memory
- Learning
- Decision

MACHINE

- Data
- Storage
- Training
- Prediction

Both learn from past experiences to make better **future decisions**.

Data = Information = Knowledge ?

Why Traditional Programming Fails

TP approach

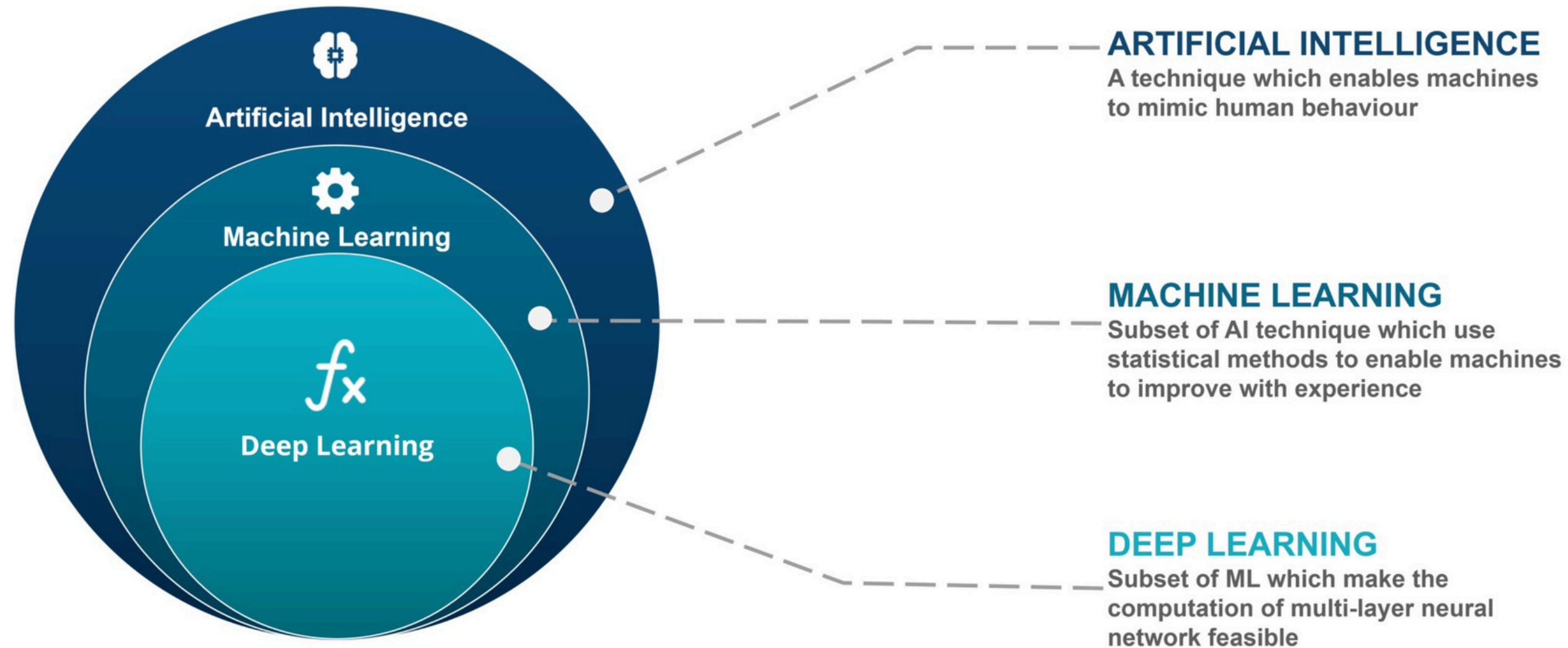
Data + Rules → Output

- Too Many Rules
- Too Many Exceptions
- Too Many Possibilities

```
1 def is_spam(email):
2     if "free" in email.lower():
3         return "Spam"
4     return "Not Spam"
5
6 print(is_spam("Free lottery! Claim now"))
7 print(is_spam("Free food in college canteen"))
```

Can machines learn the rules automatically?

Machine Learning



Machine Learning enables computers to learn patterns from data and improve their performance without being explicitly programmed.

How Machine Learning helps in Pattern finding?

Pattern 1:

[2, 4, 6, 8, 10, 12, 14, 16]

What pattern do you observe?

Pattern 2:

[124, 127, 132, 139, 148, 159, 172, 187, 204, 223, 244, 267, 292, 319, 348, 379, 412, 447, 484, 523, 564, 607, 652, 699, 748, 799, 852, 907, 964, 1023, 1084, 1147, 1212, 1279, 1348, 1419, 1492, 1567, 1644, 1723, 1804, 1887, 1972, 2059, 2148, 2239, 2332, 2427, 2524, 2623]

Can you still find the pattern?

ML approach

Data + Output → Rules

Humans are good at simple patterns.

Machines help discover patterns hidden in massive data.

Type of Machine Learning

Machine Learning

```
graph TD; ML[Machine Learning] --> SL[supervised learning]; ML --> UL[unsupervised learning]; ML --> RL[reinforcement learning];
```

supervised learning

- The model trains on labeled data.
- Predicts discrete categories or distinct classes.
- Linear Regression, Decision Trees, Support Vector Machines (SVM).

unsupervised learning

- The model trains on unlabeled data.
- Predicts discrete categories or distinct classes.
- Linear Regression, Decision Trees, Support Vector Machines (SVM).

reinforcement learning

- The model trains on unlabeled data.
- Predicts discrete categories or distinct classes.
- Linear Regression, Decision Trees, Support Vector Machines (SVM).

INTERACTIVE DISCUSSION

**“Can machines truly UNDERSTAND, or are they just very good at
PATTERN MATCHING?”**



Thank You